

TECHBIZ SECURITY

INTERNSHIP REPORT

WEEK 2 - TASK 6

VULNERABLE LAB SETUP - METASPLOITABLE 2

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1. Introduction

This report covers Week 2, Task 6 of the Ethical Hacking (EH) domain of the SOC internship at TechBiz Security. The task involved setting up a vulnerable lab environment using Kali Linux as the attacking/testing machine and Metasploitable 2 as the intentionally vulnerable target machine. Intentionally vulnerable systems such as Metasploitable 2 are widely used in cybersecurity training because they allow practitioners to practice scanning, enumeration, and exploitation techniques in a legal and controlled environment, without the risks associated with testing against live production systems.

This report documents the process of configuring both virtual machines on the same network, identifying their assigned IP addresses, and verifying connectivity between them, which together form the foundation required for any further vulnerability assessment or penetration testing work.

2. Objective

- Set up Metasploitable 2 as a vulnerable target machine.
- Configure Kali Linux and Metasploitable 2 for network communication.
- Place both systems on the same network.
- Identify the IP address of the target machine.
- Verify connectivity between Kali Linux and Metasploitable 2.
- Prepare a safe environment for future security testing and vulnerability assessment activities.

3. Lab Environment

The lab was built using Oracle VirtualBox as the virtualisation platform, with both virtual machines configured to use a Bridged Adapter so that they would receive IP addresses on the same subnet as the host network. The environment used for this task is summarised below.

Component	Details
Attacker / Testing Machine	Kali Linux
Target Machine	Metasploitable 2
Virtualization Platform	Oracle VirtualBox
Network Configuration	Bridged Adapter
Kali Linux IP Address	10.206.175.48/24
Metasploitable 2 IP Address	10.206.175.43/24

Since both machines were configured with the Bridged Adapter setting, they obtained IP addresses within the same subnet, which allowed direct network communication between them for the purposes of this lab exercise.

4. Metasploitable 2 Setup

An existing Metasploitable 2 virtual machine was used as the intentionally vulnerable target environment for this task. Metasploitable 2 is a deliberately insecure Linux-based virtual machine that is purpose-built for cybersecurity training and controlled security testing. It contains a range of

unpatched services and known vulnerabilities, making it a suitable and safe target for practicing vulnerability assessment and exploitation techniques in an isolated lab setting.

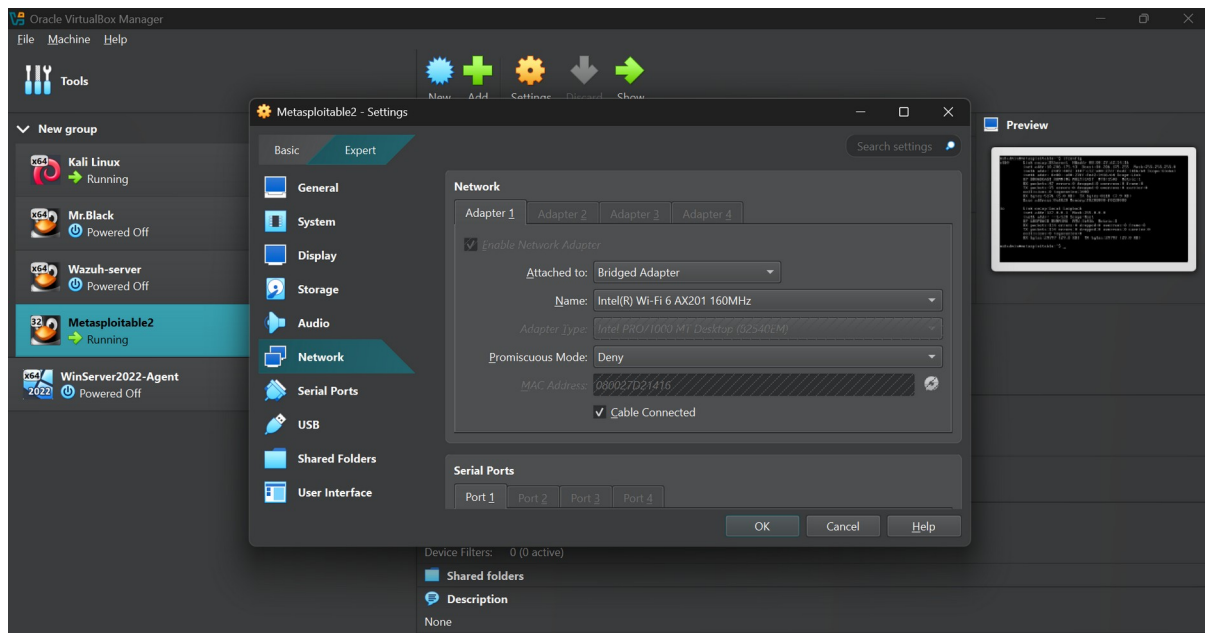


Figure 1: Metasploitable 2 VirtualBox Network Configuration

5. Kali Linux Network Configuration

The network adapter settings for the Kali Linux virtual machine were configured within VirtualBox, using a Bridged Adapter to place the machine on the same network as the target. This configuration allowed Kali Linux to obtain an IP address directly from the local network, enabling it to communicate with other devices on that network, including the Metasploitable 2 target machine.

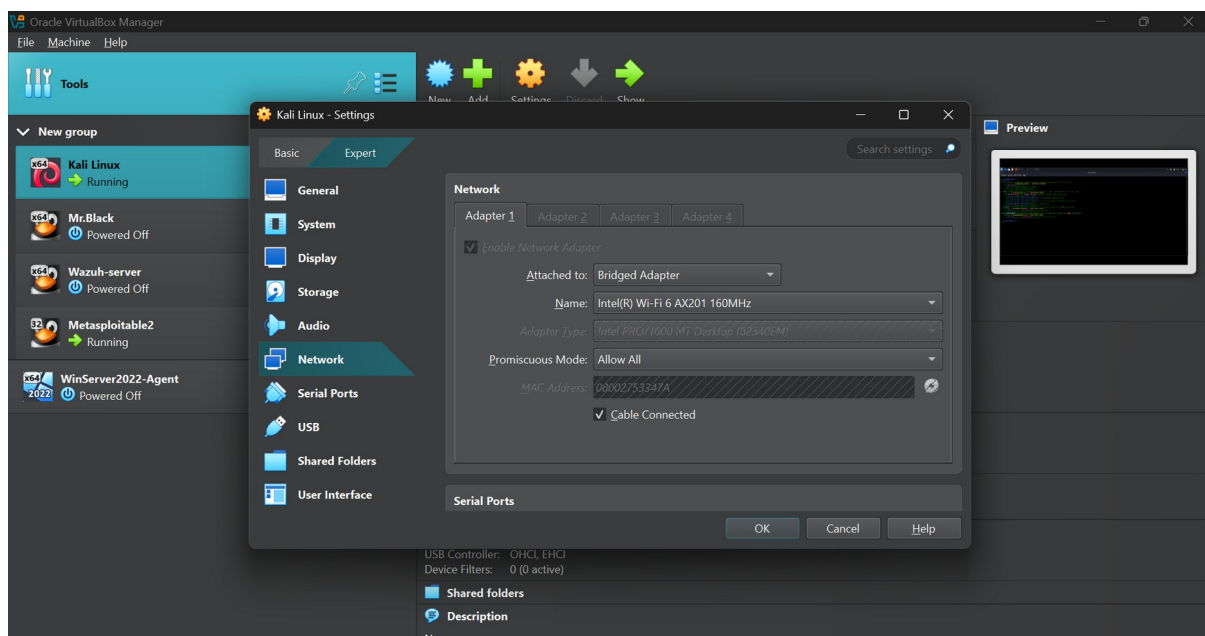
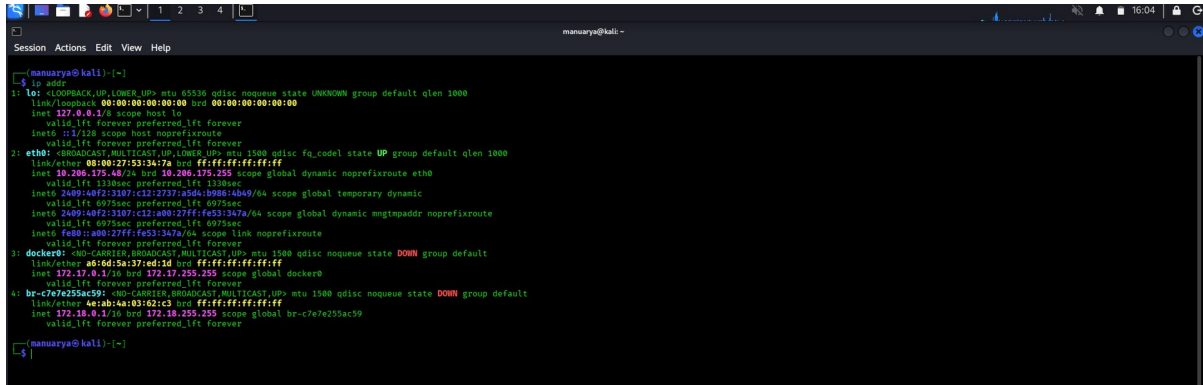


Figure 2: Kali Linux VirtualBox Network Configuration

6. IP Address Verification - Kali Linux

The Kali Linux interface configuration was checked using the `ip addr` command. The output confirmed that the `eth0` interface was assigned the IP address `10.206.175.48/24`, confirming that the machine had successfully obtained a network address on the bridged network.



```
manuara@kali:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:53:34:7a brd ff:ff:ff:ff:ff:ff
    inet 10.206.175.48/24 scope global dynamic noprefixroute eth0
        valid_lft 1388sec preferred_lft 1388sec
    inet6 2409:40f2:3107:c12:a00:27ff:fe53:347a/64 scope global dynamic temporary dynamic
        valid_lft 6075sec preferred_lft 6075sec
    inet6 2409:40f2:3107:c12:a00:27ff:fe53:347a/64 scope global dynamic mngtaddr noprefixroute
        valid_lft 6075sec preferred_lft 6075sec
    inet6 fe80::a00:27ff:fe53:347a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether a6:6d:5a:37:ed:1d brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 scope global docker0
        valid_lft forever preferred_lft forever
4: br-c7e7e255ac59: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 4e:a3:ba:03:02:c3 brd ff:ff:ff:ff:ff:ff
    inet 172.18.0.1/16 brd 172.18.255.255 scope global br-c7e7e255ac59
        valid_lft forever preferred_lft forever
manuara@kali:~$
```

Figure 3: Kali Linux IP Address Configuration

7. IP Address Verification - Metasploitable 2

The network configuration of the Metasploitable 2 machine was checked using the `ifconfig` command. The output confirmed that the `eth0` interface on the target machine was assigned the IP address `10.206.175.43`, with a subnet mask of `255.255.255.0`, placing it on the same /24 network as the Kali Linux machine.

```
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:d2:14:16
          inet addr:10.206.175.43  Bcast:10.206.175.255  Mask:255.255.255.0
          inet6 addr: 2409:40f2:3107:c12:a00:27ff:fed2:1416/64 Scope:Global
          inet6 addr: fe80::a00:27ff:fed2:1416/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:47 errors:0 dropped:0 overruns:0 frame:0
          TX packets:75 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:5176 (5.0 KB)  TX bytes:8118 (7.9 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:114 errors:0 dropped:0 overruns:0 frame:0
          TX packets:114 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:29797 (29.0 KB)  TX bytes:29797 (29.0 KB)

msfadmin@metasploitable:~$
```

Figure 4: Metasploitable 2 IP Address Configuration

8. Connectivity Verification

Connectivity between Kali Linux and Metasploitable 2 was verified by sending ICMP echo requests from Kali Linux to the Metasploitable 2 IP address, `10.206.175.43`, using the `ping` command. The test transmitted 4 packets, all 4 of which were received successfully, resulting in 0% packet loss.

This successful response confirms that Kali Linux is able to communicate with the Metasploitable 2 target over the configured bridged network, which validates that the lab environment is correctly set up for further security testing activities.

```

manuara@kali:~$ ping -c 4 10.206.175.43
PING 10.206.175.43 (10.206.175.43) 56(84) bytes of data:
64 bytes from 10.206.175.43: icmp_seq=1 ttl=64 time=7.50 ms
64 bytes from 10.206.175.43: icmp_seq=2 ttl=64 time=3.48 ms
64 bytes from 10.206.175.43: icmp_seq=3 ttl=64 time=3.50 ms
64 bytes from 10.206.175.43: icmp_seq=4 ttl=64 time=5.41 ms

--- 10.206.175.43 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 384ms
rtt min/avg/max/mdev = 3.499/5.486/7.563/1.437 ms
manuara@kali:~$

```

Figure 5: Successful Connectivity Test from Kali Linux to Metasploitable 2

9. Network Configuration Summary

Parameter	Value
Kali Linux	10.206.175.48/24
Metasploitable 2	10.206.175.43/24
Network Type	Bridged Adapter
Connectivity Status	Successful
Packet Loss	0%

10. Security and Ethical Considerations

Metasploitable 2 is an intentionally vulnerable machine and should only be used within an authorised, isolated, or otherwise controlled cybersecurity laboratory environment. Because it contains multiple unpatched and exploitable services, it should never be connected to an untrusted network or the internet, as doing so could expose it to compromise by unauthorised parties and potentially allow it to be used as a pivot point into other systems. Keeping the lab isolated and used strictly for authorised training purposes ensures that all testing activity remains safe, legal, and contained.

11. Practical Learning Outcomes

- Virtual machine network configuration.
- Basic network troubleshooting.
- IP address identification.
- Subnet awareness.
- Connectivity verification.
- Understanding attacker and target lab architecture.
- Preparing a controlled environment for ethical hacking and vulnerability assessment.

12. Conclusion

Week 2, Task 6 involved successfully configuring a vulnerable lab environment using Kali Linux and Metasploitable 2. Both systems were placed on the same network using a Bridged Adapter configuration, their IP addresses were identified as 10.206.175.48/24 for Kali Linux and 10.206.175.43/24 for Metasploitable 2, and connectivity between them was verified with a successful ping test showing 0% packet loss. This environment is now ready to be used for authorised cybersecurity training and future vulnerability assessment exercises.